

Dioxins, polychlorinated biphenyls, methyl mercury and omega-3 polyunsaturated fatty acids as biomarkers of fish consumption.



BACKGROUND/OBJECTIVES: To assess biomarkers and frequency questions as measures of fish consumption. SUBJECTS/METHODS: Participants in the Fishermen substudy numbered 125 men and 139 women (aged 22-74), and in the Health 2000 substudy, 577 men and 712 women (aged 45-74) participated. The aim of the Fishermen study was to examine the overall health effect of fish consumption in a high-consumption population, whereas the aim of the Health 2000 substudy was to obtain in-depth information on cardiovascular diseases and diabetes. Fish consumption was measured by the same validated food frequency questionnaire (FFQ) in both the studies, with a further two separate frequency questions used in the Fishermen substudy. Dioxins, polychlorinated biphenyls (PCBs) and methyl mercury (MeHg) (in the Fishermen substudy alone), and omega-3 polyunsaturated fatty acids (omega-3 PUFAs) (in both studies) were analyzed from fasting serum/blood samples. RESULTS: The Spearman's correlation coefficients between FFQ fish consumption and dioxins, PCBs, MeHg and omega-3 PUFAs were respectively 0.46, 0.48, 0.43 and 0.38 among the Fishermen substudy men, and 0.28, 0.36, 0.45 and 0.31 among women. Similar correlation coefficients were observed between FFQ fish consumption and serum omega-3 PUFAs in the Health 2000 substudy, and also between FFQ fish consumption and the frequency questions on fish consumption in the Fishermen substudy. According to multiple regression modeling and LMG metrics, the most important fish consumption biomarkers were dioxins and PCBs among the men and MeHg among the women. CONCLUSIONS: Environmental contaminants seemed to be slightly better fish consumption biomarkers than omega-3 PUFAs in the Baltic Sea area. The separate frequency questions measured fish consumption equally well when compared with the FFQ.